

Exponential graphs and transformations



1 Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

- Sketch the three functions on the same set of axes.
- Determine whether each of the following statements are true:
 - None of the curves cross the x -axis.
 - They all have the same y -intercept.
 - All of the curves pass through the point $(1, 2)$.
 - All of the curves have a maximum value.
- State the y -intercept of each curve.

2 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

- Can the value of y ever be zero or positive? Explain your answer.
- Is $y = -(2^x)$ an increasing or decreasing function?
- Describe the behaviour of the function as x increases.
- State the domain.
- State the range.

3 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = -3^x$:

- $(5, \quad)$
- $\left(\quad, -\frac{1}{27} \right)$
- $(-1, \quad)$
- $(\quad, -81)$

4 Determine the y -intercept of all exponential functions of the form:

- $y = a^x$
- $y = a^{-x}$
- $y = -a^x$
- $y = -a^{-x}$

5 Consider the function $y = -2.5 \times 4^x$.

- Is $y = -2.5 \times 4^x$ an increasing or decreasing function?
- As x approaches $-\infty$, what value does y approach?
- As x approaches ∞ , what value does y approach?
- State the y -intercept of the graph.



6 Determine whether following are increasing or decreasing exponential functions:

a $y = 10 \times \left(\frac{3}{5}\right)^x$

b $y = 9 \times 3^x$

c $y = 3 \times (0.5)^x$

d $y = 0.2 \times 2^x$

e $y = 2.2 \times 1.05^x$

f $y = 50 \times \left(\frac{7}{4}\right)^x$

g $y = 500 \times 0.75^x$

h $y = 80 \times 2^{-x}$

i $y = -5 \times 2^x$

j $y = -0.5 \times 0.8^x$

k $y = -2 \times \left(\frac{1}{4}\right)^x$

l $y = -4 \times 3^{-x}$

7 Which of the following graphs of exponential functions rises most steeply?

A $y = 2 \times (2.1)^x$

B $y = 2 \times (2.2)^x$

C $y = 2 \times (1.7)^x$

D $y = 2 \times (1.2)^x$

8 Of the two functions $y = 2^x$ and $y = 3 \times 2^x$, which is increasing more rapidly for $x > 0$?

9 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

a Find the y -intercept of $y = 2^x$.

b Hence, determine the y -intercept of $y = 2^x - 2$.

c State the horizontal asymptote of $y = 2^x$.

d Hence, determine the horizontal asymptote of $y = 2^x - 2$.

10 For each of the following exponential functions:

i Find the y -value of the y -intercept.

ii Determine the horizontal asymptote.

iii Sketch the graph of the function.

a $y = 3^{-x} - 1$

b $y = -3^x - 2$

11 For each of the following exponential functions:

i Find the y -value of the y -intercept.

ii State the domain.

iii State the range.

iv Sketch the graph of the function.

a $y = 3^x - 5$

b $y = 4^x + 3$

12 For each of the following exponential functions:



- i Find the y -value of the y -intercept.
- ii Create a table of values for $-3 \leq x \leq 3$.
- iii Determine the horizontal asymptote.
- iv Sketch the graph of the function.
- iv Is the function increasing or decreasing?

a $y = -3^{-x}$

b $y = 3^{2x}$

c $y = 3 \times 2^x + 2$

13 Consider the function $y = 8^{-x} + 6$.

- a What value is 8^{-x} always greater than?
- b Hence, what value is $8^{-x} + 6$ always greater than?
- c How many x -intercepts does $y = 8^{-x} + 6$ have?
- d State the equation of the asymptote of the curve $y = 8^{-x} + 6$.
- e State the domain.
- f State the range.

14 Consider the function $y = 2(3^x)$.

- a Find the y -value of the y -intercept.
- b Can the function value ever be negative? Explain your answer.
- c State the domain.
- d State the range.
- e As x approaches positive infinity, what value does y approach?
- f Sketch the graph of the function.

15 Consider the function $y = -10^x$.

- a Complete the following table:

x	-2	-1	0	1	2	3
y						

- b For any value of x , is y always negative for this equation? Explain your answer.
- c Sketch the graph of the function.

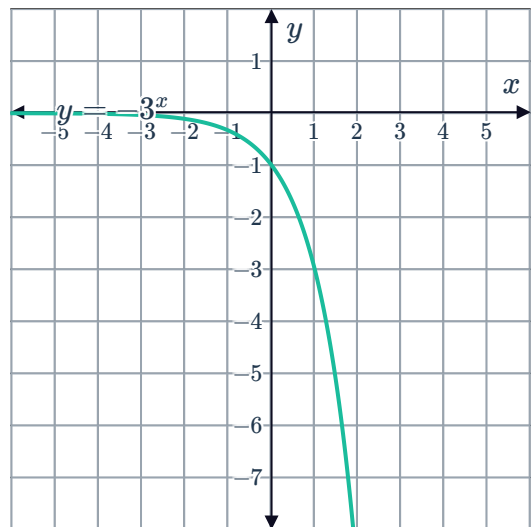
16 Consider a graph of $y = 6^{3x}$.



- a A horizontal dilation by what factor takes the graph of $y = 6^x$ to $y = 6^{3x}$?
- b Determine whether the following functions are equivalent to $y = 6^{3x}$:
 - i $y = 216 \times 6^x$
 - ii $y = \frac{1}{3} \times 6^x$
 - iii $y = 18^x$
 - iv $y = 216^x$
- c Sketch the graph of $y = 6^{3x}$.

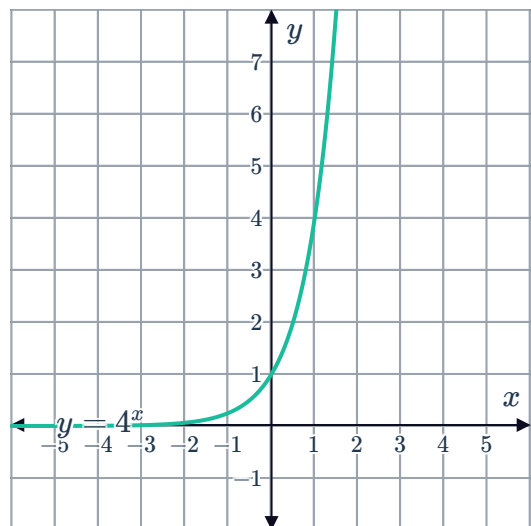
17 Consider the given graph of $y = -3^x$:

- a State the asymptote of $y = -3^x$.
- b Hence, find the asymptote of $y = 2 - 3^x$.
- c How many x -intercepts would $y = 2 - 3^x$ have?
- d State the domain of $y = 2 - 3^x$.
- e State the range of $y = 2 - 3^x$.



18 Consider the given graph of $y = 4^x$:

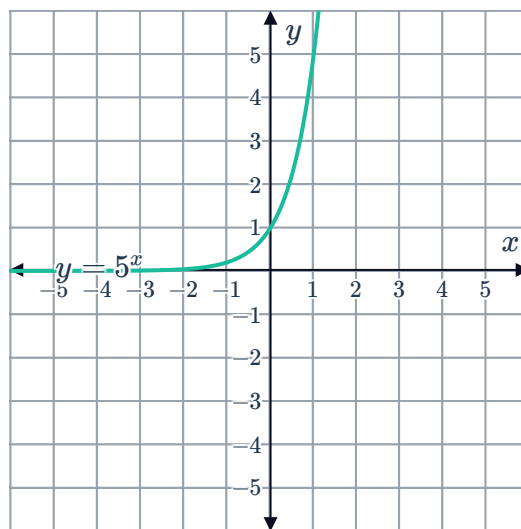
- a Describe a transformation of the graph $y = 4^x$ that would obtain $y = 4^{-x}$.
- b Sketch the graph of $y = 4^{-x}$.



19 Consider the given graph of $y = 5^x$:

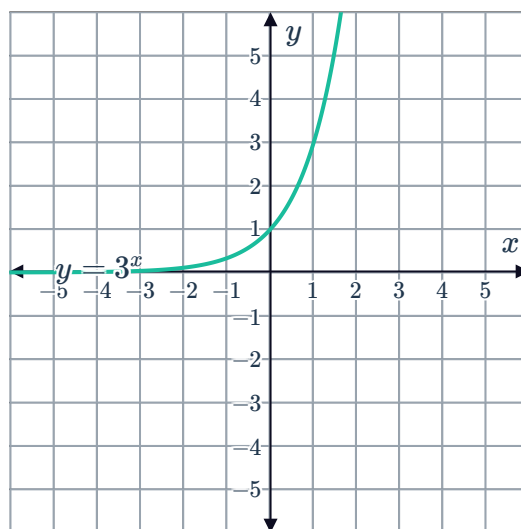


- a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$:
- b Sketch the graph of $y = -5^x$.



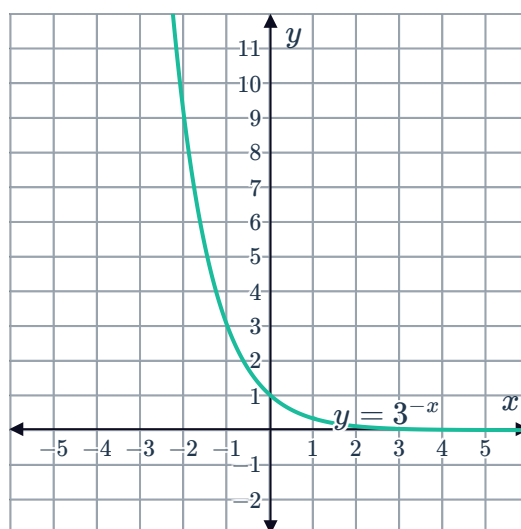
20 Consider the given graph of $y = 3^x$:

- a Describe a transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.
- b Sketch the graph of $y = 3^x - 4$.



21 Consider the given graph of $y = 3^{-x}$:

- a Describe a transformation of the graph of $y = 3^{-x}$ that would obtain $y = 3^{-x} + 2$.
- b Sketch the graph of $y = 3^{-x} + 2$.

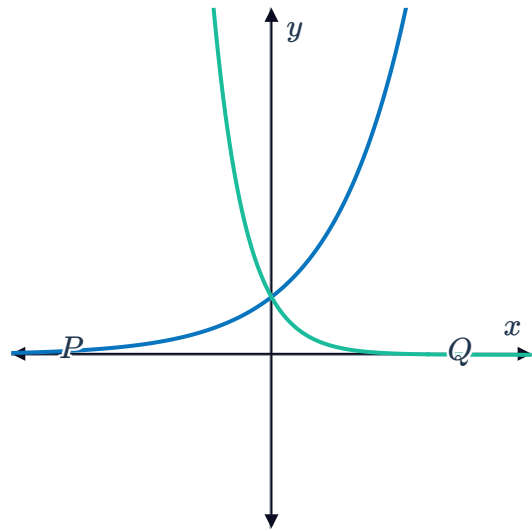




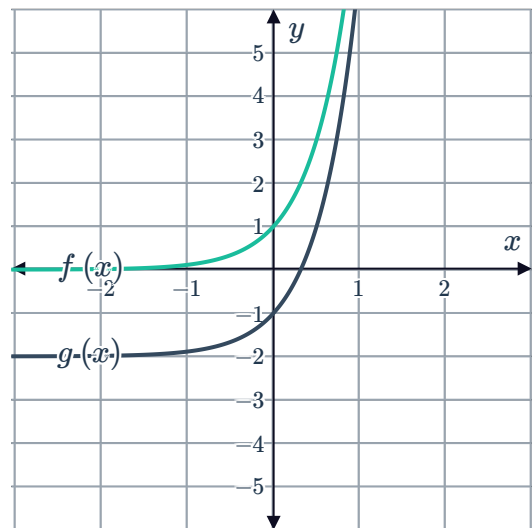
Function from a graph

- 22** Consider the given graphs of the two exponential functions P and Q :
State whether the following pairs of equations could be the equations of the graphs P and Q :

- | | |
|--|--|
| a $P : y = 2^x$
$Q : y = 2^{-x}$ | b $P : y = (3.5)^x$
$Q : y = 6^{-x}$ |
| c $P : y = 2^x$
$Q : y = 5^{-x}$ | d $P : y = 5^x$
$Q : y = 2^{-x}$ |



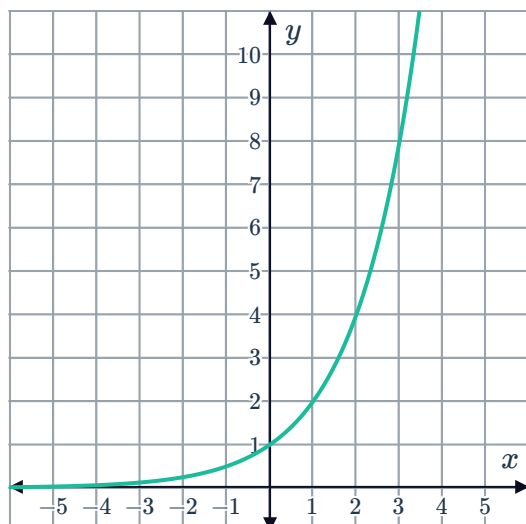
- 23** The graph of $f(x) = 9^x$ and another exponential function, $g(x)$ is shown:
 $g(x)$ increasing at exactly the same rate as $f(x)$, but has a different y -intercept. Write down the equation of function $g(x)$.



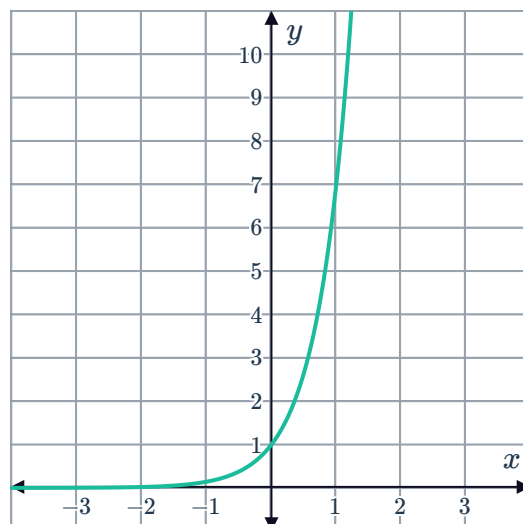
24 For each of the following graphs of exponential functions in the form $y = a^x$, state the equation of the function:



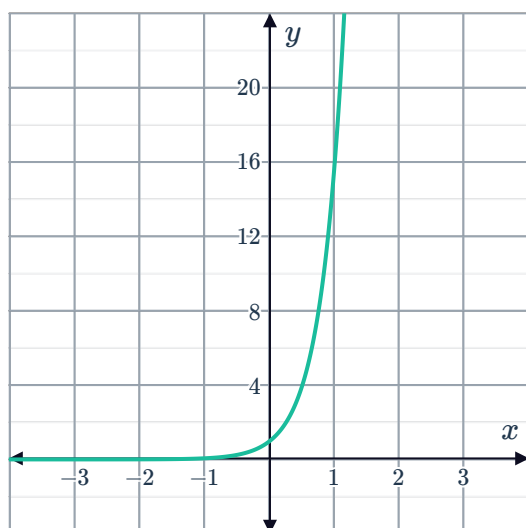
a



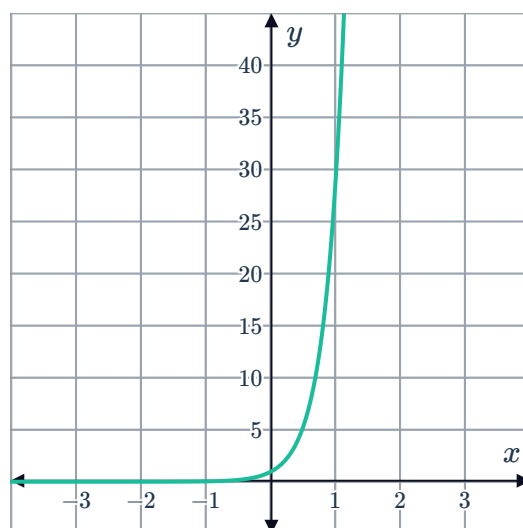
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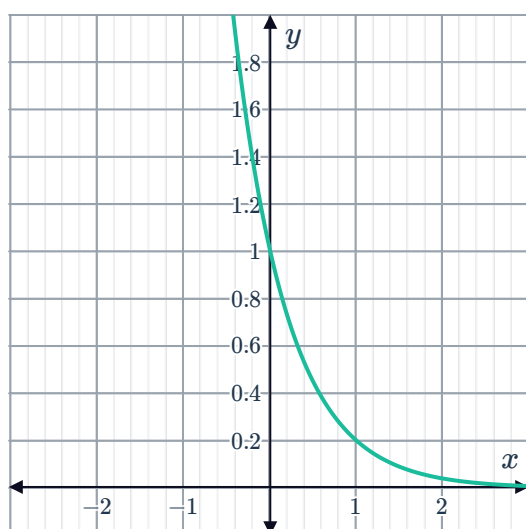
c



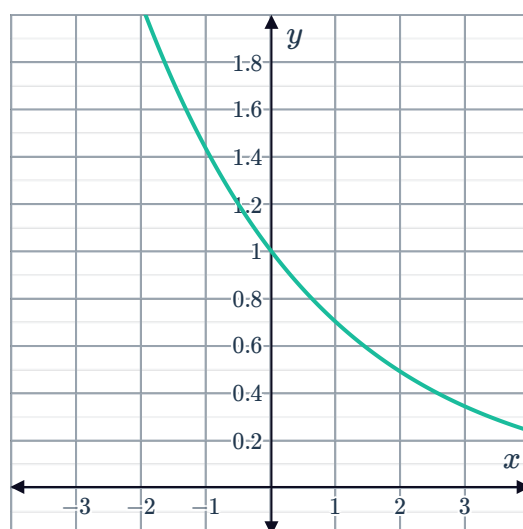
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e



f

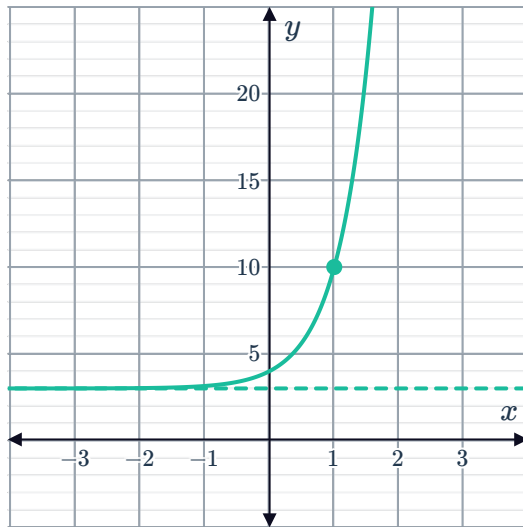


25 For each of the following graphs of exponential functions in the form $y = a^x + k$:

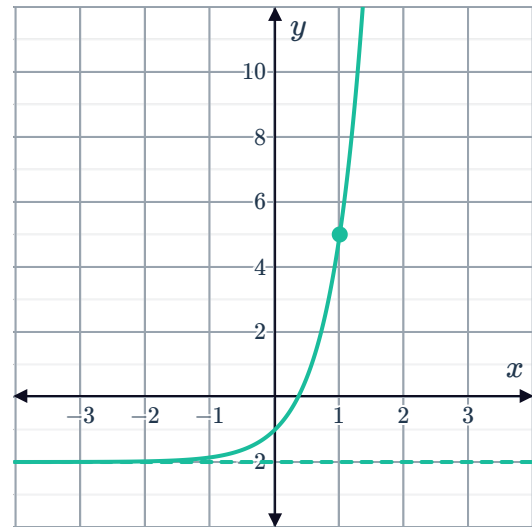


- i State the equation for the horizontal asymptote.
- ii State the equation for the exponential function.

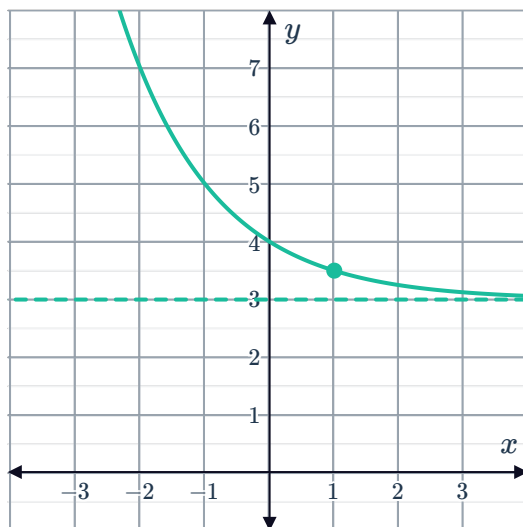
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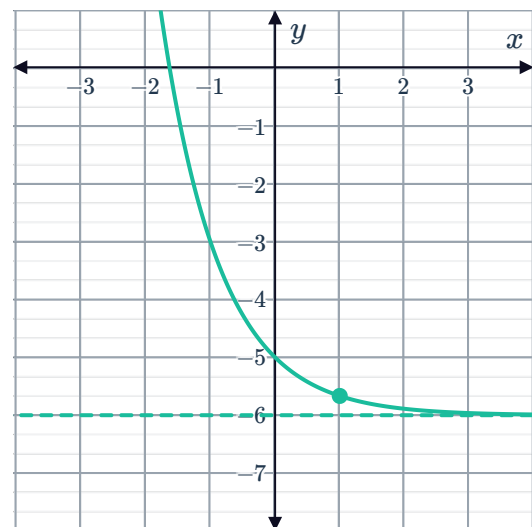
b



c



d



26 For each of the given pair of points, find the equation of the exponential function of the form $y = A \times a^x$ that passes through the points:

- | | |
|------------------------------------|---------------------------------------|
| a $P(0, 3)$ and $Q(1, 6)$ | b $P(1, 30)$ and $Q(2, 90)$ |
| c $P(1, 20)$ and $Q(2, 5)$ | d $P(1, 30)$ and $Q(2, 10)$ |
| e $P(2, 20)$ and $Q(3, 40)$ | f $P(2, -18)$ and $Q(5, -486)$ |